

- 1(a) A researcher plans to carry out a statistical investigation to test whether there is linear correlation between the time (T weeks) from conception to birth, and the birth weight (W grams) of new-born babies.

Explain why a 1-tail test is appropriate in this context.

[1]

- (b) The researcher records the values of T and W for a random sample of 11 babies. They calculate Pearson's product-moment correlation coefficient for the sample and find that the value is 0.722.

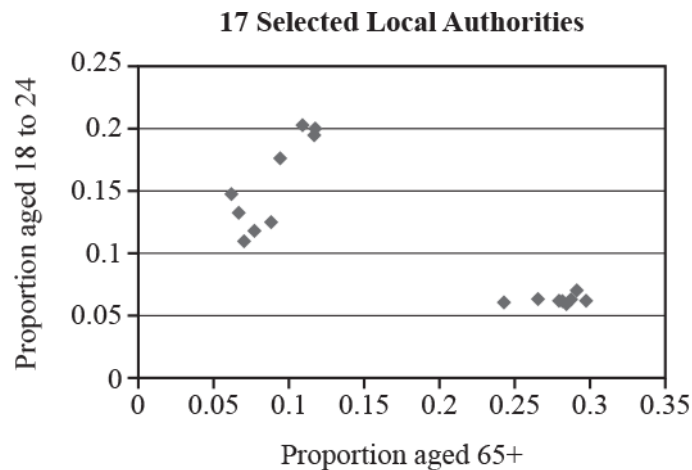
Use the table below to carry out the test at the 1% significance level.

Critical values of Pearson's product-moment correlation coefficient.

	1-tail test	5%	2.5%	1%	0.5%
	2-tail test	10%	5%	2.5%	1%
n	10	0.5494	0.6319	0.7155	0.7646
	11	0.5214	0.6021	0.6851	0.7348
	12	0.4973	0.5760	0.6581	0.7079
	13	0.4762	0.5529	0.6339	0.6835

[5]

2(a) A trainer was asked to give a lecture on population profiles in different Local Authorities (LAs) in the UK. Using data from the 2011 census, he created the following scatter diagram for 17 selected LAs.



He selected the 17 LAs using the following method. The proportions of people aged 18 to 24 and aged 65+ in any Local Authority are denoted by P_{young} and P_{senior} respectively. The trainer used a spreadsheet to calculate the

value of $k = \frac{P_{\text{young}}}{P_{\text{senior}}}$ for each of the 348 LAs in the UK. He then used specific ranges of values of k to select the 17 LAs.

Estimate the ranges of values of k that he used to select these 17 LAs. [2]

(b) Using the 17 LAs the trainer carried out a hypothesis test with the following hypotheses.

H_0 : There is no linear correlation in the population between P_{young} and P_{senior} .

H_1 : There is negative linear correlation in the population between P_{young} and P_{senior} .

He found that the value of Pearson's product-moment correlation coefficient for the 17 LAs is - 0.797, correct to 3 significant figures.

(i) Use the Critical Values table in part d) below to show that this value is significant at the 1% level. [2]

(ii) The trainer concluded that there is evidence of negative linear correlation between P_{young} and P_{senior} in the population.

Use the scatter diagram from part a) to comment on the reliability of this conclusion. [2]

(c) Describe one outstanding feature of the population in the areas represented by the points in the bottom right hand corner of the diagram. [1]

(d) The trainer's audience included representatives from several universities.

Suggest a reason why the diagram might be of particular interest to these people.

[1]

Critical values of Pearson's product-moment correlation coefficient

1-tail test	5%	2.5%	1%	0.5%
2-tail test	10%	5%	2%	1%
<i>n</i>				
1	-	-	-	-
2	-	-	-	-
3	0.9877	0.9969	0.9995	0.9999
4	0.9000	0.9500	0.9800	0.9900
5	0.8054	0.8783	0.9343	0.9587
6	0.7293	0.8114	0.8822	0.9172
7	0.6694	0.7545	0.8329	0.8745
8	0.6215	0.7067	0.7887	0.8343
9	0.5822	0.6664	0.7498	0.7977
10	0.5494	0.6319	0.7155	0.7646
11	0.5214	0.6021	0.6851	0.7348
12	0.4973	0.5760	0.6581	0.7079
13	0.4762	0.5529	0.6339	0.6835
14	0.4575	0.5324	0.6120	0.6614
15	0.4409	0.5140	0.5923	0.6411
16	0.4259	0.4973	0.5742	0.6226
17	0.4124	0.4821	0.5577	0.6055
18	0.4000	0.4683	0.5425	0.5897
19	0.3887	0.4555	0.5285	0.5751
20	0.3783	0.4438	0.5155	0.5614
21	0.3687	0.4329	0.5034	0.5487
22	0.3598	0.4227	0.4921	0.5368
23	0.3515	0.4132	0.4815	0.5256
24	0.3438	0.4044	0.4716	0.5151
25	0.3365	0.3961	0.4622	0.5052
26	0.3297	0.3882	0.4534	0.4958
27	0.3233	0.3809	0.4451	0.4869
28	0.3172	0.3739	0.4372	0.4785
29	0.3115	0.3673	0.4297	0.4705
30	0.3061	0.3610	0.4226	0.4629

3 Laxmi wishes to test whether there is linear correlation between the mass and the height of adult males.

(a) State, with a reason, whether Laxmi should use a 1-tail or a 2-tail test. [1]

Laxmi chooses a random sample of 40 adult males and calculates Pearson's product-moment correlation coefficient, r . She finds that $r = 0.2705$.

(b) Use the table below to carry out the test at the 5% significance level. [5]

Critical values of Pearson's product-moment correlation coefficient.

	1-tail test	5%	2.5%	1%	0.5%
	2-tail test	10%	5%	2.5%	1%
n	38	0.2709	0.3202	0.3760	0.4128
	39	0.2673	0.3160	0.3712	0.4076
	40	0.2638	0.3120	0.3665	0.4026
	41	0.2605	0.3081	0.3621	0.3978

4 In an experiment involving a bivariate distribution (X, Y) a random sample of 7 pairs of values was obtained and Pearson's product-moment correlation coefficient r was calculated for these values.

- (a) The value of r was found to be 0.894. Use the table below to test, at the 5% significance level, whether there is positive linear correlation in the population, stating your hypotheses and conclusion clearly. [5]

1-tail test	5%	2.5%	1%	0.5%
2-tail test	10%	5%	2%	1%
n				
1	—	—	—	—
2	—	—	—	—
3	0.9877	0.9969	0.9995	0.9999
4	0.9000	0.9500	0.9800	0.9900
5	0.8054	0.8783	0.9343	0.9587
6	0.7293	0.8114	0.8822	0.9587
7	0.6694	0.7545	0.8329	0.9745
8	0.6215	0.7067	0.7887	0.8343
9	0.5882	0.6664	0.7498	0.7977
10	0.5494	0.6319	0.7155	0.7646

Scatter diagrams for four sets of bivariate data, are shown.

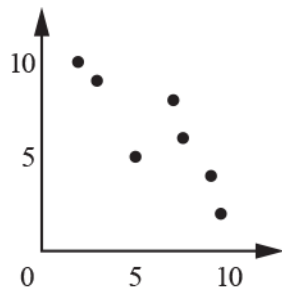


Diagram A

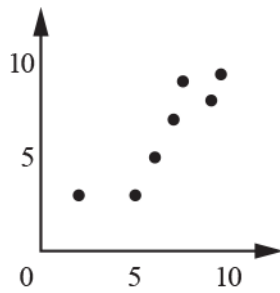


Diagram B

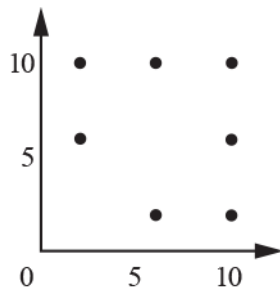


Diagram C

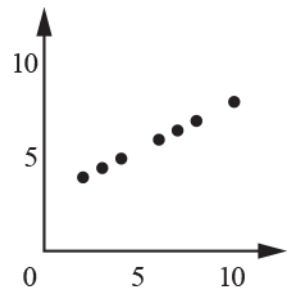


Diagram D

It is given that $r = 0.894$ for **one** of these diagrams.

(b) For each of the **other** diagrams, state how you can tell that $r \neq 0.894$.

[3]

- 5 Maria planned a statistical investigation into trees of a certain variety. She wished to test whether there is positive linear correlation between the height of a tree and the circumference of its trunk at the base.

(a) State, with a reason, whether a 1-tail or a 2-tail test is more appropriate.

[1]

Maria recorded the height and circumference of a random sample of 10 trees of this variety in a wood near her

home. She calculated the product-moment correlation coefficient for her sample and found that the value was 0.642.

(b) Use the table below to carry out the test at the 2.5% significance level.

[5]

(c) Give two reasons why it would not be appropriate to use Maria's results to draw a conclusion about all trees of this variety.

[2]

Critical values of Pearson's product-moment correlation coefficient.

	1-tail test	5%	2.5%	1%	0.5%
	2-tail test	10%	5%	2.5%	1%
n	9	0.5822	0.6664	0.7498	0.7977
	10	0.5494	0.6319	0.7155	0.7646
	11	0.5214	0.6021	0.6851	0.7348
	12	0.4973	0.5760	0.6581	0.7079

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		Very likely weight will increase with time oe or He is only looking for positive correlation	B1 [1]	Or eg "Expect weight to increase with time" oe "Foetuses grow" oe Ignore all else <u>Examiner's Comments</u> Many candidates gave a general answer, perhaps learned by rote: "Because the researcher is looking for either an increase or a decrease, not both." This does not reflect the actual context of the question, so did not gain the mark.
	b		$H^0: \rho = 0$ Allow other letters $H^1: \rho > 0$ where ρ is the correlation coefficient for the population For hypotheses in words, not using parameter: or where ρ is the correlation coefficient between time and weight Comp 0.722 with 0.6851 Reject H^0 . Condone Accept H^1 There is evidence of (positive linear) correlation between time from conception to birth and weight of new-born babies Or eg It appears that birth weight increases with time (from conception to birth)	B1 B1 M1 M1 A1 [5]	B1B0 for 1 error, eg undefined ? or 2-tail For hypotheses in words, not using parameter: H^0 : There is no correlation between time and weight H^1 : There is positive correlation between time and weight B1B0 But omission of "positive": B0B0 May be implied by conclusion Allow without "positive" and without "linear" In context, not definite <u>Examiner's Comments</u>  Misconception One difficult feature of hypothesis testing is that in some cases the calculated value of the statistic is significant when it is greater than the critical value, whereas in other cases the calculated value is significant when it is less than the critical value. In both these questions there was confusion about this.
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
2	a		$k > 1.4$ $k < 0.25$ or $1.4 < k < \dots\dots$ $\dots\dots < k < 0.25$ (allow $k > 1.1$ to 1.6) (allow $k < 0.2$ to 0.3) (ranges as above) B1 B1	B1(AO2.2b) B1(AO2.2b) [2]	Allow $\geq$ and $\leq$ SC: $0.25 < k < 1.4$: B1B0 (ranges as on left) <u>Examiner's Comments</u> Centres are referred to the general comment on Question 11 under "Section A Overview" above. For the types of answers that were acceptable in Question 11 parts (b)(ii), (c) and (d), centres are referred to the published mark scheme. Few candidates answered this correctly. Most candidates did not understand which pairs of proportions to divide in order to find the limiting values of k. A helpful approach would be to draw a line from the origin, just meeting the point in a cluster nearest to the line $y = x$. This point provides the correct pair of proportions for one of the limiting values of k. Then repeat for the other cluster.	Allow "x"
	b	i	$0.797 > 0.5577$ or $-0.797 < -0.5577$ or $-0.797 > 0.5577$	B2(AO3.1b) (AO3.2a) [2]	$0.797 > 0.6055$ or $-0.797 < -0.6055$ B1 ± 0.5577 B1 <u>Examiner's Comments</u> Most candidates answered this correctly. Two common errors were "$-0.797 < 0.5577$, hence significant" and $-0.797 < -0.6055$ (i.e. a two-tail test).	Allow $\geq$ or $\leq$

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
		ii	There are clusters (or groups etc.) Apparent good correlation caused by clusters or Two clusters with no –ve corr'n within them or a comment similar to one of the above. AND Conclusion unreliable or Value of r is misleading oe	B1*(AO2.3) B1dep B1*(AO3.5b) [2]	or Not bivariate normal distribution B1 so use of tables for r not valid B1 <u>Examiner's Comments</u> Many candidates thought the statement was reliable, because the diagram does show negative correlation. Many other candidates said the conclusion was not reliable because the diagram omits data from a large number of LAs. A few noted the key point that the points on the diagram are in two clusters. But not many explained that this results in an apparent (but false) appearance of good correlation. A few candidates noted, correctly, that the points cannot be contained roughly within an ellipse and therefore the distribution was probably not bivariate normal, so the use of Pearson's coefficient is invalid.	NOT Too scattered Not represent whole pop Small sample Clusters not on reg line B1B0
	c		High prop of 65+ or Low prop of 18-24 Prop of young very similar, or ≈ 0.06 Proportion of senior to young is high	B1(AO2.2b) [1]	If consider only <u>one</u> age-group, must be proportion not number If consider <u>both</u> age-groups, allow eg Higher number of seniors than young or Many seniors, few young <u>Examiner's Comments</u> Many candidates gave good answers. Some, incorrectly, stated that the diagram showed that relevant LAs had a large <u>number</u> of older residents.	NOT: Similar proportions of 65+ Population is elderly

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
	d		Top left points contain high prop of 18-24s. (So these LAs may be areas where there are universities or where they can recruit)	B1(AO2.2b) [1]	Shows places where large nos of 18-24s Shows where to focus recruiting. So universities can recruit. 18-24s are their target group. No need to specify "Top left group"	Allow "students" or "young" instead of "18-24s" Any implication that diagram enables you to see information about location of young people
			Total	8		
3		a	1-tail, because expect mass and height to be positively correlated	B1 (AO 1.1) [1]	1-tail because generally expect taller people to be heavier	or equivalent
		b	$H_0: \rho = 0$ $H_1: \rho > 0$ where ρ is pmcc for population comp 0.2638 Reject H_0 There is evidence that the mass and height of adult males are positively correlated	B1 (AO 1.1) B1 (AO 2.5) B1 (AO 1.1) M1 (AO 2.2b) A1 (AO 3.5a) [5]	Omit definition of ρ : B1B0	
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
4		a	H_0: There is no linear correlation between X and Y H_1: There is positive linear correlation between X and Y Compare with 0.6694 Reject H_0 There is evidence of positive linear correlation between X & Y	B1 (AO1.1) B1 (AO2.5) B1 (AO1.1) M1 (AO1.1) A1 (AO2.2b) [2]	B1B0 for one error, eg omission of "linear" OR "+ve" In context, not definite	Or $p = 0$ Or $p > 0$
		b	A: Negative (linear) correlation C: No (linear) correlation D: $r = 1$	E1 (AO1.2) E1 (AO2.2a) E1 (AO1.2) [3]	or points not close to straight line, or $r = 0$ Not "r small" or "poor correlation"	Allow without "linear" Allow without "linear"
			Total	8		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5		a	1-tail. Testing for "positive" linear correlation	E1(AO 3.1b) [1]	or Expect larger circumference to go with larger height oe	
		b	$H_0: \rho = 0$ $H_1: \rho > 0$ where ρ is lin correlation coeff in pop Comp 0.642 with 0.6319 Reject H_0 . There is evidence of +ve (linear) corr'n between height & circ of trees of this variety (in this wood)	B1(AO1. 1) B1(AO2. 5) M1(AO1. 1) M1(AO1. 1) A1(AO2. 2b) [5]	B1B0 for 1 error, eg undefined ρ or 2-tail In context, not definite.	Allow omission of "linear" throughout Allow without "linear" and / or "in this wood"
		c	eg sample is small conditions in other areas may be different	E1(AO 3.5a) E1(AO 3.5a) [2]		
			Total	8		